

OSSP (Operating Systems And Systems Programming) - 25CS2104E

Roll No. : 2520030029

Name:Pujith Pavan Kumar

SEC-8

SKILL-1:

Aim: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parentchild relationships, inspect process tree, practice system call tracing

```
siddinapujith07@DESKTOP-FDHCAIT:~$ cd ~
siddinapujith07@DESKTOP-FDHCAIT:~$ mkdir LinuxShellProject
siddinapujith07@DESKTOP-FDHCAIT:~$ cd LinuxShellProject
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ mkdir src include
build docs tests
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ touch Makefile README.md
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ touch src/main.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ ls -R
.:
Makefile  README.md  build  docs  include  src  tests
./build:
./docs:
./include:
./src:
main.c
./tests:
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ nano src/main.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ nano Makefile
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ make
gcc -Wall src/main.c -o shell
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ ./shell
Linux Shell Project Started!
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ ps
  PID TTY          TIME CMD
   339 pts/0    00:00:00 bash
   633 pts/0    00:00:00 ps
```

```
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ pstree
systemd--agetty
          |chornyd-starter--chornyd--chornyd
          |cron
          |dbus-daemon
          |buit-systemd(Ub)---SessionLeader---Relay(339)---bash---+
++
          |   |init---{init}
          |   |login---bash
          |   |{init-systemd(Ub)}
          |--networkd-dispat
          |--rsyslogd---3*[{rsyslogd}]
          |--systemd---sd-pam
          |--systemd-journal
          |--systemd-logind
          |--systemd-resolve
          |--systemd-udev
          |--unattended-upgr---{unattended-upgr}
          |--wsl-pro-service---7*[{wsl-pro-service}]
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ nano src/fork_demo.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ gcc src/fork_demo.c -o fork_demo
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ ./fork_demo
Hello
Hello
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ nano src/exec_demo.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ gcc src/exec_demo.c -o exec_demo
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ ./exec_demo
Makefile  build  exec_demo  include  src
README.md  docs  fork_demo  shell  tests
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ nano src/parent_child.c
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ gcc src/parent_child.c -o parent_child
siddinapujith07@DESKTOP-FDHCAIT:~/LinuxShellProject$ ./parent_child
I am Parent Process
I am Child Process
```

```
venkatsaikrishna5@VSK-HP:~/LinuxShellProject$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffd95dfef6e0 /* 26 vars */) = 0
brk(NULL) = 0x62d2e39ea000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x779fa2423000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x779fa241e000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779fa23eb000
mmap(0x779fa23f2000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x779fa23f2000
mmap(0x779fa2413000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779fa2413000
mmap(0x779fa241a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x779fa241a000
mmap(0x779fa241c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x779fa241c000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779fa23bd000
mmap(0x779fa23c1000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x779fa23c1000
mmap(0x779fa23e5000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779fa23e5000
mmap(0x779fa23e9000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x779fa23e9000
```

```
|VtO|FFO|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...} = 0
brk(0x62d2e3a4d000) = 0x62d2e3a4d000
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VtO|FFO|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...} = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_yp
ixel=0}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_yp
ixel=0}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VtO|FFO|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...} = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VtO|FFO|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOH|IEXTEN|ECHOCTL|ECHOKE, ...} = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
openat(AT_FDCWD, ".", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY
) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
newfstatat(AT_FDCWD, ".", {st_mode=S_IFDIR|0755, st_size=4096,
..., 0}) = 0
getdents64(3, 0x62d2e3a1e150 /* 13 entries */, 32768) = 384
getdents64(3, 0x62d2e3a1e150 /* 0 entries */, 32768) = 0
close(3) = 0
write(1, "Makefile build exec_demo inc"... , 100Makefile b
uild exec_demo include shell tests
README.md docs fork_demo parent_child src
) = 100
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NU
LL) = 0
munmap(0x779fa2420000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
```

Code:

```
#include <stdio.h>

int main()
{
    printf("Linux Shell Project Started!\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    printf("Hello\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if(pid == 0)
        printf("I am Child Process\n");
    else
        printf("I am Parent Process\n");

    return 0;
}
```

```
#include <unistd.h>

int main()
{
    execl("/bin/ls", "ls", NULL);
    return 0;
}
```